

Kenan Firmansyah

Game Developer & AI Researcher

realkenanfir@gmail.com | +62 851-7544-7540 | Jakarta, Indonesia
nandev.dev | LinkedIn | GitHub

Professional Summary

Game developer and Computer Engineering graduate. I build agent-based AI for VR/XR and gameplay in Unreal Engine and Unity, mostly using Fuzzy Logic, Behavior Trees, and ML. On my own time I am researching long-term memory and personality consistency for autonomous AI agents.

Education

Bachelor of Computer Engineering (Game Technology) 2020 – 2024

Electronic Engineering Polytechnic Institute of Surabaya (EEPIS), Indonesia

- **Thesis/Final Project:** *Xanthous* – A VR horror game built in Unreal Engine with layered NPC AI. Implemented Fuzzy Logic, Monte Carlo Simulation, FSMs, and Behavior Trees to drive agent perception and decision-making.

Technical Skills

Game Engines: Unreal Engine 4/5, Unity
Programming: C++, C#, Python, Swift
AI & ML: Reinforcement Learning (Q-Learning), Fuzzy Logic, Behavior Trees, Monte Carlo, LLM Integration, PyTorch
Infrastructure: Git, Docker, CI/CD, Firebase, GCP
Mobile & Web: iOS SDK (SwiftUI, CoreML, WatchKit, SpriteKit), Flutter
Data: Scraping (Selenium), Analysis, Visualization (Tableau, Gephi)

Licenses & Certifications

Unity Certified Associate: Programmer Sep 2024 – Sep 2027

Issued by Unity Technologies | Verify on Credly

Selected Publications

- **Adaptive Radio System Using MetaSounds for Serious VR Game.** *IEEE Access* (2025). Built adaptive audio in Unreal Engine using State Machines and MetaSounds.
- **Real Time Procedural Audio for Immersive Gameplay with MetaSounds.** *TCCE* (2025). Generated environmental audio (rain, wind, thunder) procedurally in Unreal Engine, cutting storage needs while keeping realistic output.
- **Dynamic Level of Difficulties Using Q-Learning and Fuzzy Logic.** *IEEE Access* (2024). Q-Learning + Fuzzy Logic for real-time difficulty adjustment in a management sim.
- **Enhancing Serious Game Experience Through In-Game Radio Using Context-Aware Recommender System.** *INASS IJIES* (2024). Deep learning recommender that adapts to in-game context in real time.
- **Dynamic Day and Night Cycle Impact in a Serious VR Game.** *ICVR Conference* (2024).

Academic & Independent Research

Clone World: AI Agents Personality & Memory Research Ongoing

Lead AI Researcher / Developer

- Building a life simulation where AI agents remember past interactions and shift personality based on what happens to them.
- Using Hexagonal Architecture so the system can run on any engine.
- Wiring in local LLMs so agents can hold conversations and generate their own story threads.

- Fisherman Manager: Dynamic Difficulty Adjustment (DDA) Research**2024
- Lead Programmer & Designer
- Built a management simulation in Unreal Engine 5 as a testbed for dynamic difficulty adjustment research.
 - Combined Q-Learning with Fuzzy Logic to read player behavior and adjust game state variables in real time.

Professional Experience

Game DeveloperOngoing

Agate, Indonesia

- Working on core gameplay systems for an unannounced title using the studio's engineering pipeline.

Lead DeveloperJul 2025 – Present

JoyBait Studio, Indonesia

- Running a team of 8 developers in an indie studio spun off from the Apple Developer Academy.
- Leading development on *Saturated*, coordinating between roles and writing gameplay code.

Game Developer (Unity) & QAMar – May 2025

Ecosoft Interactive, Remote

- Handled iOS development and QA testing for a mobile arcade project.
- Found and fixed logic bugs in the core gameplay loop; improved UI responsiveness.
- Set up testing protocols and build schedules for a remote team.

Apple OS DeveloperFeb – Desember 2025

Apple Developer Academy @ BINUS, Bali, Indonesia

- Completed a 10-month program building native iOS and WatchOS applications from scratch.
- Used CoreML for on-device image analysis (*Tona*, a photo grading app) and integrated LLMs for semantic matching (*Findect*, a networking app).

Lead Developer (Unreal Engine)2024 – February 2025

Starpixel Studio, Remote

- Built gameplay and system mechanics for an undisclosed multiplayer horror game.
- Wrote multiplayer synchronization and AI replication logic for low-latency client interactions.

Data Scientist (Intern)2023

PT. eBdesk, Indonesia

- Wrote Python scripts with Selenium to scrape and clean web data at scale.
- Analyzed and visualized datasets using Gephi and Tableau for B2B client reports.

Selected Achievements

- **1st Place Winner, iPlay Game Jam 2025:** System Developer & Enemy AI Programmer (Apple Developer Academy).
- **Google Play x Unity Certification:** Completed 7-month technical training in game programming and architecture.
- **Game Jams (10+):** Shipped playable builds in ScoreSpace, GMTK, Brackeys, and others, using Unity, UE, and Photon multiplayer.